

Reputation Scheme as a Monitoring Device ^{*}

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Abstract

We experimentally investigate how reputation schemes promote cooperation in trust dilemmas by separating their dual functions: monitoring and information provision. We novelly employ a two-stage design to disentangle the reputation-recording phase from the reputation-revealing phase and compare two types of schemes: a centralized system that records full reputation and a decentralized system that records partial reputation. We find that both partial and full reputation schemes enhance cooperation compared to the case where no reputation scheme is in place. However, the positive effects operate primarily through monitoring: cooperation emerges strongly when partners' actions are recorded, but collapses when monitoring ceases. Our analysis reveals strategic reputation building as the key behavioral mechanism: over half of partners achieve perfect cooperation records during monitoring, only to systematically exploit initiators' trust once monitoring ceases. This highlights the importance of continuous monitoring in sustaining cooperative behavior.

Keywords: Reputation mechanisms; trust game; experiments.

JEL Classification: C72; D82; C92.

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1 Introduction

In the race to seek collaborations, firms increasingly face a critical strategic dilemma: collaborate with external partners to accelerate success or protect proprietary knowledge from potential exploitation. When pharmaceutical giant Merck partnered with biotech firm Moderna to develop cancer therapies in 2018, this dilemma came sharply into focus. Their collaboration promised higher success rates—research shows open innovation partnerships in pharmaceuticals can triple the probability of successful drug development¹—yet required Merck to share proprietary research methods, compound data, and screening technologies that Moderna could potentially appropriate after establishing the joint collaboration. Such risks are well-documented, as studies show confidential information frequently leaks through innovation networks (Alberti & Pizzurno, 2017) and proprietary technologies can be systematically copied or misappropriated during external collaborations (Cohen, Nelson, & Walsh, 2000). This tension between value creation through openness and the risk of opportunistic behavior from partners represents the fundamental trust dilemma in open innovation (Chesbrough, 2003; Chesbrough, Vanhaverbeke, & West, 2006; Foege, Lauritzen, Tietze, & Salge, 2019; Stefan, Hurmelinna-Laukkanen, Vanhaverbeke, & Oikarinen, 2022).

This trust dilemma constitutes a pervasive barrier in countless open collaborations for services, ideas, and technologies across industries, especially in the rapidly expanding era of the gig economy and digital platforms (Nambisan, Siegel, & Kenney, 2018; Sundararajan, 2017). The Microsoft-Sendo case (Curwen, 2003) dramatically illustrates the devastating consequences of such opportunism in partnership. After Sendo shared proprietary designs under their partnership agreement, Microsoft allegedly misappropriated this manufacturing know-how and redirected it to Sendo's competitors. The sequential nature of interactions and asymmetric information created the one-sided incentive problem (Grossman & Hart, 1986; Williamson, 1975) and limited Sendo's ability to detect and deter this opportunistic behavior—by the time Sendo discovered the misappropriation, irreparable damage had occurred, ultimately leading to its insolvency. This pattern repeats across various collaboration contexts. Anticipating exploitation from potential partners, firms will reject valuable collaborative opportunities from the outset,

¹The result is based on Deloitte (2015)'s analysis of 281 biopharmaceutical companies during 1988-2012. Similar results were found in other industries. For instance, Harvard Business Review studies on Procter & Gamble's strategy of open innovation (Connect + Develop Program) show that it contributed to more than 35% of the company's innovations, and their innovation success rate more than doubled after implementing open innovation practices.

opting instead for inefficient internal development. This defensive posture produces substantial societal welfare losses—innovation stagnates when complementary knowledge remains siloed, R&D efforts are wastefully duplicated across organizations, and promising technologies requiring multi-firm capabilities remain undeveloped.

Reputation schemes, which come in both centralized and decentralized forms, offer promising solutions to these trust dilemmas. Centralized reputation schemes represent that a single entity aggregates and disseminates information, such as platform feedback systems (G. Bolton, Greiner, & Ockenfels, 2013) and credit bureaus (Brown & Zehnder, 2010). These formal systems coexist with decentralized alternatives such as word-of-mouth networks and third-party gossip (Fehr & Sutter, 2019; Fonseca & Peters, 2018), which distribute reputation information through informal social channels. Both formats have demonstrated substantial effectiveness across diverse economic contexts with asymmetric information. In e-commerce, seller rating systems significantly increase trade efficiency. For instance, Resnick, Zeckhauser, Swanson, and Lockwood (2006) documented a price premium of approximately 8% for highly-rated sellers in controlled experiments, while Cabral and Hortacsu (2010) found that a seller’s first negative feedback shifted their weekly sales growth from positive 5-7% to negative 7-8%, highlighting a substantial market penalty for reputation damage.² Similarly, reputation schemes effectively screen workers in labor markets (Bartling, Fehr, & Schmidt, 2012) and dramatically increase repayment rates in microfinance (De Janvry, McIntosh, & Sadoulet, 2010).

Despite robust evidence of the effectiveness of reputation schemes in various domains, a critical gap remains in our understanding of the underlying mechanisms through which they foster cooperation. Our study addresses this gap by examining how reputation schemes mitigate trust dilemmas in open innovation settings. Specifically, we distinguish two fundamental functions of reputation schemes, each operating in a different phase. The first function, *monitoring*, operates during the *reputation-recording phase*, potentially deterring opportunistic behavior as partners strive to build favorable reputations. The second function, *information*, operates during the *reputation-revealing phase* by disclosing information about partners’ past actions, thereby enabling the identification of trustworthy collaborators. In practice, these functions are deeply intertwined, as reputation information is simultaneously constructed and utilized through repeated interactions to

²Recent surveys on reputation and feedback systems in online markets include G. E. Bolton and Ockenfels (2012); Gutt, Neumann, Zimmermann, Kundisch, and Chen (2019); Tadelis (2016); ?.

support cooperation (Dellarocas, 2006; Mailath & Samuelson, 2006; Wilson, 1985). This intertwining creates a significant identification challenge: when collaboration succeeds under a reputation scheme, is it because the scheme deters opportunistic behavior through monitoring concerns, or because it helps identify inherently trustworthy partners?

We answer these questions through a laboratory experiment designed to mirror the trust dilemma in open innovation. Our experimental design modifies the standard trust game (Berg, Dickhaut, & McCabe, 1995) by adding outcome uncertainty inherent to innovation contexts. We refer to the first mover as an “initiator” (she) and the second mover as a “partner” (he). The initiator chooses between closed and open innovation for a project, where the latter increases the success rate but exposes her to the potential exploitation by the partner. If open innovation is chosen, the partner subsequently decides whether to keep knowledge confidential or leak it for personal gain. This captures the essence of the scenario we described earlier - while openness enhances innovation potential, it simultaneously creates vulnerability to opportunistic behavior. The initiator must balance the potential benefits of collaboration against the risk of misappropriation, while the partner faces the temptation to exploit the initiator.

Our experimental design uniquely identifies the distinct monitoring and informational functions of reputation schemes in the reputation recording and revealing phases. Initiators interact with different partners over eight rounds divided into two stages of four rounds each. In Stage I, the reputation-recording phase, partners establish reputations that remain hidden from initiators, enabling us to identify the monitoring effect. Partners know that their actions will become visible in the future, but initiators cannot yet access this information. In Stage II, the reputation-revealing phase, initiators gain access to previously generated reputation information before making decisions, allowing us to examine the informational effect. This methodological separation provides a clear identification of how each function of a reputation scheme independently affects cooperation, addressing a critical gap in the literature on reputation system design.

We implement three reputation conditions—No Reputation, Partial Reputation, and Full Reputation—which vary in how partners’ historical behaviors from Stage I are recorded and revealed to initiators in Stage II. This between-subject variation creates controlled environments that differ only in the completeness of reputation information recording and revelation, allowing us to measure the causal effects of a reputation mechanism on facilitating cooperation. The Partial Repu-

tation condition reflects the fragmented nature of reputation information in a decentralized reputation scheme,³ whereas the Full Reputation condition provides an estimate of the effect of a centralized review system. By comparing outcomes across these three conditions, we determine not only whether reputation schemes promote cooperation, but also how different structures influence the effectiveness of both monitoring and informational functions.

We find reputation schemes significantly enhance cooperation but operate primarily through the monitoring function. Both Partial and Full Reputation schemes significantly increase cooperation rates compared to the No Reputation, with no statistical difference between the two reputation conditions—suggesting that even decentralized reputation schemes can achieve comparable benefits to centralized ones. This cooperative effect manifests strongly during Stage I, when reputation is being recorded, but deteriorates sharply in Stage II, when monitoring ceases despite the availability of reputation information. We identify strategic reputation building as the driving behavioral mechanism. Partners deliberately cultivate favorable reputations while being monitored (over half achieving perfect cooperation records), but subsequently exploit initiators' trust once monitoring ends. This pattern fundamentally undermines the informational value of reputation. We conclude that reputation schemes primarily serve as monitoring devices that temporarily alter opportunistic behavior, rather than as informational devices to identify inherently trustworthy collaborators. This challenges conventional wisdom about reputation and carries key implications for institutional design in settings with one-sided incentive problems.

Our study contributes to the experimental literature on reputation and trust by separating the reputation generation (monitoring) phase from the reputation consumption (information) phase. Prior studies have primarily focused on how different information contents affect trust and trustworthiness, examining how various types of historical information influence cooperation (G. Bolton, Katok, & Ockenfels, 2005; Charness, Du, & Yang, 2011), how endogenous information transmission shapes reputation formation (Abraham, Grimm, Neeß, & Seebauer, 2016; Fonseca & Peters, 2018), and whether reputation acts as a positive or negative influence (Grosskopf & Sarin, 2010). However, existing experimental designs typically deploy reputation schemes where monitoring and information provision

³For example, Hannigan, Seidel, and Yakis-Douglas (2018) note that rumors about product innovation in technology blogs affect firms' open innovation decisions, while Alberti and Pizzurno (2017) identify four distinct types of knowledge flow in open innovation networks within an aerospace cluster.

occur simultaneously, making it difficult to examine their distinct effects. Moreover, they may lead researchers to overestimate the informational function while underappreciating the role of active monitoring in contributing to cooperative outcomes. Our study addresses this limitation through a novel two-stage design that explicitly disentangles these functions by temporally separating the recording of reputation from its revelation. Combined with our three reputation treatments, this approach enables identification of causal effects and assessment of the relative contributions of monitoring versus information provision to cooperative behavior—distinctions that previous experimental designs could not capture.

While our finding that both partial and full reputation schemes effectively promote cooperation aligns with previous literature (Bracht & Feltovich, 2009; Keser, 2003), our discovery that these effects operate primarily through monitoring rather than information provision challenges conventional wisdom about reputation mechanisms. We demonstrate that the effectiveness of a reputation scheme in mitigating trust dilemmas depends critically on its temporal implementation—specifically, whether monitoring remains continuous throughout interactions. Our findings on how strategic reputation building mitigates the value of reputation information further extend the literature on reputation dynamics and manipulation. Partners' strategic behavior across the reputation recording and revealing phases aligns with theoretical predictions by Mailath and Samuelson (2001) and empirical findings in online markets by Cabral and Hortacsu (2010), but advances these insights by demonstrating how people strategically adapt their behavior in response to temporal changes in reputation scheme structure (Bohnet & Huck, 2004). Our findings reveal that when people anticipate transitions in reputation schemes—either from monitored to non-monitored phases or changes in monitoring intensity—the informational value of reputation is significantly weakened because people strategically build favorable reputations solely to exploit them after the transition occurs.

The rest of the paper is structured as follows: Section 2 introduces the theoretical framework for the dual roles of reputation schemes. Section 3 presents our experimental design and procedures. Section 4 provides a detailed analysis of our findings and explanations for the mechanisms of reputation schemes. Section 5 concludes.

2 Theoretical Framework

We introduce an *Initiator-Partner* game (illustrated in Figure 1) to capture the general trust issues arising from the open innovation. The initiator moves first, choosing either to develop the innovation project internally (closed innovation) or to launch an open innovation with an external partner. Innovation is subject to uncertainty in its success rate, which increases under an open R&D paradigm. We assume that if the initiator adopts a closed innovation, the project will succeed with a probability of 30%. An open collaboration increases the success rate to 70%. However, open innovation exposes the initiator to a moral hazard problem. Once the collaboration is launched, a partner can exploit the initiator by leaking information about the shared project. The partner earns extra profits from the leakage, but the initiator is worse off than she is under a paradigm of closed innovation. Since the initiator cannot determine whether a potential partner will exploit confidential information before collaboration begins, this creates a significant adverse selection problem that may fundamentally discourage open innovation. When contemplating the innovation strategy, initiators must balance the expected gains from increased success rates against potential losses from Partners' leakage behavior. We characterize this as the "success versus leakage trade-off". The unique subgame perfect Nash equilibrium is $(Close, (Leak, Leak))$, in which the initiator plays *Close*, and the partner plays *Leak* disregarding whether the project succeeds or fails.

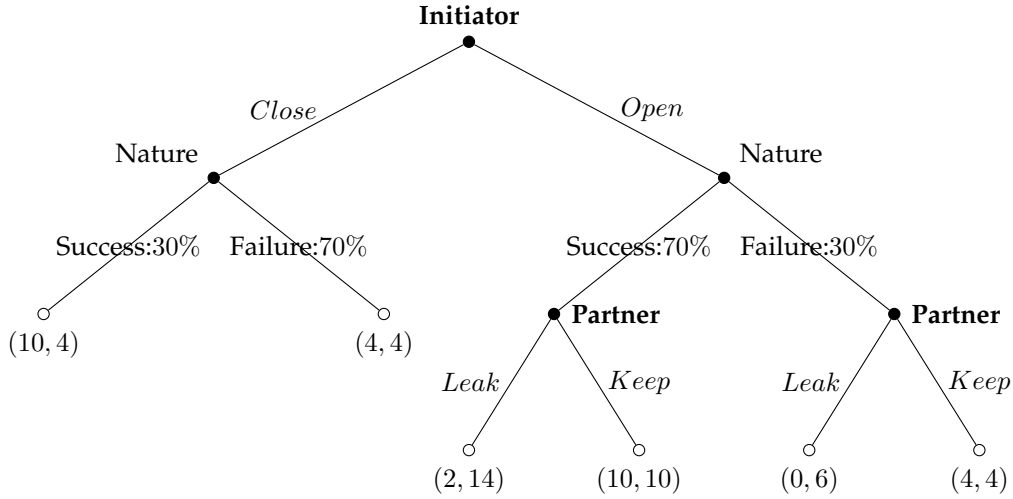


Figure 1: The game between an Initiator and a Partner

The game is a variant of Berg et al. (1995)'s trust (or investment) game, incorporating added uncertainty. The equilibrium similarly predicts uncooperative be-

havior.⁴ Our key variations are the introduction of a project associated with differential success rates between open and closed innovation strategies. We introduce them for two reasons. First, uncertainty is a prominent feature of joint collaborations involving creative activities, such as open innovations and joint research. Our “success-versus-leakage tradeoff” effectively captures the moral hazard and adverse selection problems inherent in open innovation processes. Second, this setting mitigates the certainty effect (Tversky and Kahneman (1981)), which entails the overvaluation of riskless payoffs. Note that the initiator acts before the resolution of the risk, thus evaluating the *Close* and *Open* actions based on their induced lotteries.⁵

We define a partner’s reputation as the set of their historical stage game outcomes, with a reputation scheme determining how this information is recorded and publicly revealed. As Wilson (1985) highlights, the reputation scheme creates linkages between otherwise independent stage games, thereby promoting cooperative outcomes through sustained trust incentives. Since reputations are built through sequential interactions, where actions create a history that signals trustworthiness, these linkages provide a rationale for players to deviate from the non-cooperative equilibrium and sustain cooperative actions – *Open* for initiators and *Keep* for partners. Given the potential continuation value of cooperation, partners who value future opportunities find it beneficial to keep confidentiality, while those who focus on short-run gains opt for leakage. Confronted with this heterogeneity, initiators form beliefs about which partners are likely to cooperate based on observed historical behavior. The informational function of the reputation scheme thus enables initiators to access historical outcomes and assess the probability of facing a cooperative partner. When this inferred probability is sufficiently high, initiators confidently choose open innovation. This mechanism regarding the consumption of reputation implies that when reputation schemes act as informational devices, Initiators are more likely to play *Open* with partners who have cooperative reputations.

⁴According to Johnson and Mislin (2011)’ meta survey on the standard trust game, the general finding is that trust towards strangers is common, with receivers typically returning sent amounts, though results may vary based on factors like the multiplier size, culture, and institutional development.

⁵The action *Close* leads to a simple lottery with the exogenous uncertainty (as reflected in the fixed success rate) only. The action *Open* corresponds to a lottery for which the probability distribution depends on both strategic uncertainty and exogenous uncertainty. The strategic uncertainty is reflected in the initiator’s beliefs about the Partner’s strategies. In the classic trust game (e.g., special success rates like 100% or 0%), the action *Close* yields a risk-free payoff, and choosing *Open* leads to a lottery due to strategic uncertainty.

Reputation schemes also operate as monitoring devices when they systematically record and disseminate information about agents' past behaviors. When partners' actions are observed, recorded, and made visible to potential future initiators, they face a direct causal link between current behavior and future payoff opportunities, creating a powerful deterrent against leakage. The effectiveness of this monitoring depends critically on whether the information recording process is active and ongoing. Partners make significantly different strategic calculations when they know their current choices will affect their future reputation versus when this connection is severed. When monitoring is active, choosing *Keep* becomes strategically valuable for partners interested in future collaboration opportunities. Conversely, when monitoring ends and actions no longer contribute to reputation formation, the incentive for cooperative behavior substantially weakens or disappears entirely. When reputation schemes act as monitoring devices, they create a "shadow of the future" that may temporarily transform opportunistic partners into cooperative ones, but this transformation reverses once the monitoring mechanism is removed.

Together, these informational and monitoring devices foster a positive cycle where cooperative actions build credibility, increasing both open innovation adoption and confidentiality maintenance.

3 Experimental Design and Procedures

We novelly separate the monitoring and informational function of reputation schemes by a within design. The experiment consists of two stages, where Stage I is for the reputation-recording phase, and Stage II is for the reputation-revealing phase. Each stage contains four rounds of the Initiator-Partner game. In Stage I, partners' decisions are recorded but withheld from initiators, allowing the reputation scheme to function exclusively as a monitoring device for partners.⁶ In Stage II, each initiator is informed of their currently matched partner's decisions recorded during Stage I. There is no record-keeping of the game outcomes in Stage II, and thus, a reputation scheme exclusively functions as an informational device to initiators.

⁶The partner's decision in each round will be recorded as (Close, -), (Open, Keep), or (Open, Leak). The first case indicates that the initiator has chosen closed innovation, meaning the partner is not involved in the decision-making process. The last two cases indicate that the initiator has chosen open innovation, with the partner either keeping confidentiality or leaking information for extra profit, respectively.

At the start of each session, equal numbers of subjects are randomly assigned to be initiators or partners. We used the terminology “company” and “external developer” in the instructions to represent initiators and partners, respectively. Subjects maintain fixed roles throughout the experiment, but are re-matched with different counterparts in each round. After each round, subjects can observe the outcomes of their interactions. We match initiators and partners as absolute strangers to prevent direct reciprocity. All subjects are aware that they will never play with the same participants twice.

Unlike standard reputation schemes, which typically involve fixed partners and create linkages across repeated interactions with the same counterpart (Kreps & Wilson, 1982; Milgrom & Roberts, 1982), our scheme establishes a temporal connection between Stage I and Stage II while incorporating stranger matching. In other words, the reputation mechanisms discussed by Wilson (1985)—which promote cooperative outcomes by creating linkages between otherwise independent stage games and incentivizing deviations from non-cooperative equilibria—apply to our setup as well, with the added feature of phased separation to isolate monitoring and informational functions. Consequently, our reputation scheme encourages partners to build a good reputation in Stage I by choosing *Keep*, as those who value future opportunities can capitalize on the continuation value of cooperation in Stage II, while short-run maximizers may opt for leakage.

We implement three reputation treatments that differ in the amount of reputation information to be recorded in Stage I and revealed in Stage II. These conditions are: No Reputation, Partial Reputation, and Full Reputation. We use these conditions to investigate the impact of the information volume on the reputation mechanism. Table 1 presents the treatments designed for three reputation conditions.

	No Reputation	Partial Reputation	Full Reputation
Record & reveal rounds	none	random 2 rounds	4 rounds
# subjects	64	64	64

Table 1: Comparisons of treatments across the three reputation conditions. In Stage I, the game outcomes are recorded for two or four rounds, and in Stage II, initiators are shown the partners’ actions from those recorded rounds.

No Reputation is the benchmark treatment in which the computer does not record any game outcomes, establishing a baseline where no reputations are built

in Stage I nor any reputations are disclosed in Stage II. In the *Full Reputation* treatment, the computer records all the interaction outcomes in Stage I and presents them to the matched initiators in Stage II. The Full Reputation condition maps the centralized reputation scheme, providing the initiator with complete reputation information. The *Partial Reputation* treatment mirrors the decentralized reputation scheme where the computer randomly selects only two rounds from Stage I, creating uncertainty about which actions will determine reputation.

Our treatment variations are specifically designed to identify the two functions of reputation schemes. The monitoring function is isolated by comparing partners' choices across treatments in Stage I, where game structures are identical except for partners' knowledge about how their actions translate into reputation. The information function is identified by comparing initiators' choices in Stage II, where reputation information about partners becomes available. Prior research suggests that both the comprehensiveness of monitoring and the completeness of available information should affect cooperative outcomes, leading to our main hypotheses:

Hypothesis 1 (Monitoring Function). *In Stage I, partners' Keep actions are ordered as follows: Full = Partial > Baseline.*

Hypothesis 2 (Informational Function). *In Stage II, initiators' Open actions are ordered as follows: Full > Partial > Baseline.*

These hypotheses reflect distinct expectations for the monitoring and information functions. For monitoring (Hypothesis 1), we expect comparable cooperation levels between Full and Partial treatments but significantly higher than in the baseline treatment, drawing on research that observation mechanism through a pre-stage trust game (Bracht & Feltovich, 2009) or a third neutral party (Fehr & Sutter, 2019). They point out that even a small amount of information about the allocator's most recent action and the existence of a third-party observer are sufficient to yield a substantial improvement, suggesting that both monitoring mechanisms should effectively deter partners from leaking information. For information (Hypothesis 2), we predict a graduated effect where more complete reputation histories generate greater trust, aligning with findings by Keser (2003) and Duffy, Xie, and Lee (2013) that longer reputation histories enhance market efficiency. This research suggests our Full treatment would provide initiators with more reliable information about partners' tendencies than the partial treatment, while both should exceed the baseline where no reputation information is available.

We ran the laboratory experiment at the Research Center for Behavioral Science

of Wuhan University in China with 192 participants from Bachelor’s and Master’s programs. We conducted four sessions for each treatment, with sixteen subjects in each session. Each experimental session lasted approximately one and a half hours, with participants earning an average payment of 58.8 RMB, including a show-up fee. In each round, the project outcomes were determined by drawing a crystal ball from a black bag containing ten balls in two colors: blue and red. Drawing a red ball indicated a successful project, whereas drawing a blue ball indicated failure. Participants could review historical information throughout the experiment.

4 Experimental Results

4.1 Reputation Schemes and Cooperative Outcomes

Overall treatment effects. We first pool the outcomes from Initiator-Partner games across three reputation schemes to examine whether reputation schemes promote cooperation and improve social welfare. We focus on three game outcomes (*Close*, —), (*Open*, *Keep*), and (*Open*, *Leak*), with the cooperative outcome (*Open*, *Keep*) yielding the highest expected social welfare.⁷ As shown in Table 2, the frequency of achieving the cooperative outcome (*Open*, *Keep*) is 16% under the No Reputation condition. This number rises significantly to 29% and 34% with Partial and Full Reputation schemes, respectively. Moreover, there is no significant difference in game outcomes between the Partial and Full Reputation schemes, suggesting that the weaker and incomplete reputation scheme is also effective in promoting cooperation in trust dilemma contexts when seeking partnerships.

Game Outcome	No Reputation	Partial Reputation	Full Reputation
(<i>Close</i> , —)	66%	52%**	48%**
(<i>Open</i> , <i>Keep</i>)	16%	29%**	34%**
(<i>Open</i> , <i>Leak</i>)	18%	19%	19%

Table 2: Frequencies of outcomes from Initiator-Partner games across three reputation schemes. Statistical significance is evaluated relative to the No Reputation condition. Comparisons between Partial Reputation and Full Reputation reveal no significant differences in any of the three game outcomes. Statistical tests employed are two-sample proportion test with ** $p < 0.01$.

⁷The expected social welfare is 9.8 in the case of a closed innovation, 16.4 for the cooperative outcome (*Open*, *Keep*), and 13 for the exploitative outcome (*Open*, *Leak*).

Figure 2 illustrates the distribution of the game outcomes across different treatments and stages. In Stage I, where the reputation scheme primarily serves as a monitoring device by making partners' actions visible to future initiators, the established pattern persists. Both Partial and Full Reputation treatments significantly increase the frequency of the cooperative outcome (*Open, Keep*), which rises from 18% in the No Reputation condition to 40% in the Partial scheme and 52% in the Full scheme. These results suggest that both reputation schemes improve cooperative behavior as a monitoring device.

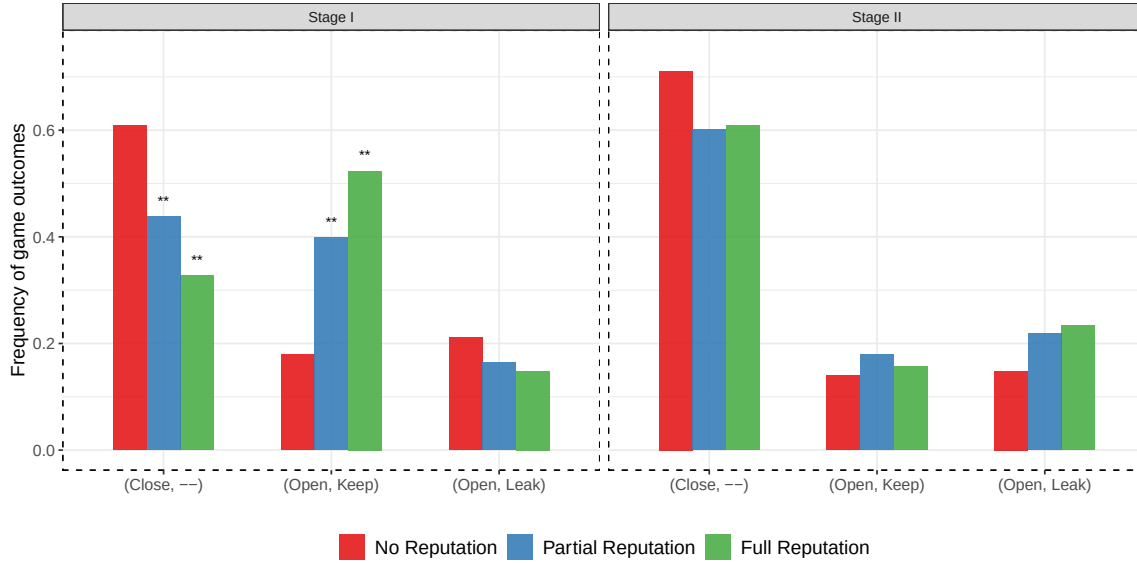


Figure 2: Histogram of Initiator-Partner game outcomes by stage. Statistical significance is evaluated relative to the No Reputation condition. Comparisons between Partial and Full Reputation treatment reveal no significant differences across all outcomes at the 5% level. Statistical tests employed are two-sample proportion test with * $p < 0.05$; ** $p < 0.01$.

In Stage II, where the monitoring function is removed and the reputation scheme primarily serves as an information device for initiators, the frequencies of the cooperative outcome under both Partial and Full Reputation treatment drop to the baseline level. This contradicts our hypothesis concerning the informational effect of reputation schemes. A comparison between the two stages reveals that, in the Partial and Full treatments, initiators shift from open to closed innovation, and partners become less likely to choose *Keep* over *Leak* once reputation monitoring ceases. The sharp decline in cooperation during Stage II highlights the importance of ongoing monitoring in sustaining cooperative behavior.

Result 1. *Both Partial and Full reputation schemes equally enhance cooperative outcomes and social welfare, primarily driven by Stage I, where the reputation schemes solely serve as monitoring devices that regulate partners' reputation-building for choosing Keep.*

We also find that the overall treatment effects remain robust, regardless of project success or failure. In Table 6 (see Appendix B.1), we consider the interactions between project outcome and partners' decisions and similarly tabulate the frequencies of the five game outcomes. Both Partial and Full Reputation treatments show similar patterns: the reputation schemes, whether partial or full, significantly enhance cooperative outcomes following both project success and failure. Figure 8 in Appendix B.1 further demonstrates that the overall treatment effects are primarily driven by outcomes in Stage I.

4.2 Monitoring versus Information: Initiator and Partner Behaviors

Treatment Effects on Initiator and Partner Decision-Making. We further investigate how reputation schemes influence the cooperative decisions made by initiators and partners. Table 3 summarizes the frequencies of initiators' *Open* (versus *Close*) decisions and partners' *Keep* (versus *Leak*) decisions across different treatments and stages.⁸ Consistent with result 1, reputation schemes significantly boost cooperative decisions during Stage I. Initiators' *Open* decisions increase from a baseline of 39% to 56% under Partial and 67% under Full Reputation treatments, while partners show even stronger treatment effects with *Keep* decisions rising from 46% (baseline) to 71% (Partial) and 78% (Full). However, these cooperation gains completely disappear in Stage II when reputation building ceases, as both initiators' *Open* rates and partners' *Keep* rates revert to baseline levels. Specifically, although initiators' *Open* rates under Partial and Full reputation schemes are approximately 10 percentage points (p.p.) higher than in the baseline condition in Stage II, these increases are not significant at the 5% level.

Comparing the two stages, cooperation drops dramatically for two reputation treatments. Initiators reduce *Open* decisions by 16 p.p. in Partial and 28 in Full Reputation conditions, while partners show even larger declines in *Keep* decisions—26 p.p. in Partial and 38 in Full. This larger decline for partners indicates that the reputation scheme's monitoring function motivates cooperation more effectively than its informational function alone.

Figure 3 displays the dynamics of cooperative choices by initiators and partners. The patterns for the No Reputation treatment are comparable to the findings

⁸As in subsection 4.1, our analyses remain robust after considering the project success or failure. See table 7 in Appendix B.1 for details.

Decision	Stage	No Reputation	Partial Reputation	Full Reputation
Initiator: <i>Open</i>	I	50 (39%)	72 (56%)**	86 (67%)**
	II	37 (29%)	51 (40%)	50 (39%)
	I vs. II		††	††
Partner: <i>Keep</i>	I	23 (46%)	51 (71%)**	67 (78%)**
	II	18 (49%)	23 (45%)	20 (40%)
	I vs. II		††	††

Table 3: Frequencies of the Initiator’s *Open* decisions and the Partner’s *Keep* decisions. Statistical significance is assessed relative to the No Reputation condition at the significant level of ** $p < 0.01$. Comparisons between Partial Reputation and Full Reputation show no significant differences in any of the outcomes at the 5% level. The differences between Stage I and Stage II are significant at the level of †† $p < 0.01$. Both the two-sample *t*-test and the Wilcoxon rank sum test produce consistent significance results.

in the trust game literature, showing a typical decline in *Open* rate across periods with a noticeable restart effect (Bohnet & Huck, 2004). In Stage I, both partial and complete reputation schemes consistently maintain higher levels of *Keep* and *Open* rates compared to the baseline. Most striking is the trajectory of partners’ *Keep* decisions, reaching 87% for Full Reputation treatment and 71% for Partial Reputation treatment by period 4. This suggests that partners in our experiment understand how reputation schemes link the two stages and recognize the strategic value of reputation building in Stage I. When their actions are recorded and visible to future Initiators in Stage II, many of them act cooperatively by foregoing the short-term gains from leakage.

Upon transitioning to Stage II, the cooperation-enhancing effects of reputation schemes substantially diminish. Initiators in reputation treatments show a gradual decline in *Open* decisions while still maintaining a slight advantage over the baseline throughout Stage II — a pattern that reflects the informational value of reputation. In contrast, partners exhibit an immediate and steep drop in *Keep* rates when entering Stage II, plunging from approximately 80% to 40%, and eventually falling below baseline levels by the final period. This sharp discontinuity reveals that partners deliberately built good reputations only to exploit initiators once their actions stopped affecting their reputation. Despite the temporary boost of trust that reputation information provides to initiators, partners’ strategic exploitation ultimately drives overall cooperation below the level in Stage I under both reputation schemes, highlighting the essential function of monitoring in sustaining cooperative outcomes.

Result 2. *Both Partial and Full reputation schemes significantly enhance cooperative de-*

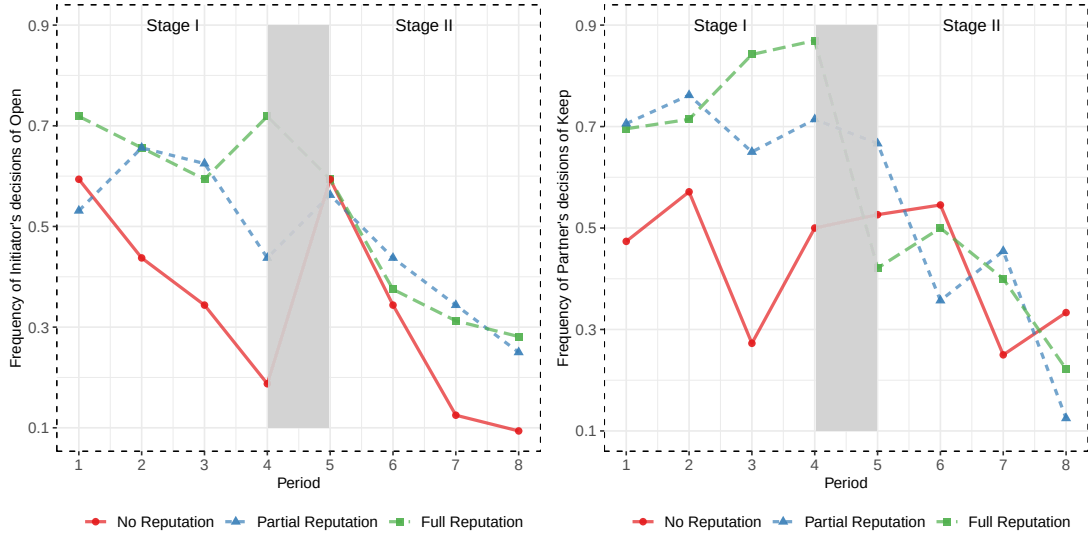


Figure 3: Frequencies of *Open* and *Keep* decisions over time. The figure tracks Initiators' *Open* decisions (left panel) and Partners' *Keep* decisions (right panel) across eight rounds. The gray bar distinguishes Stage I (periods 1-4), during which reputation is built, from Stage II (periods 5-8), where reputation information is revealed and utilized.

cisions by initiators (*Open*) and partners (*Keep*) in Stage I. However, these treatment effects disappear in Stage II, with partners exhibiting immediate and substantial declines in choosing *Keep*.

Table 4 presents linear regression models of initiator and partner decisions, providing statistical validation of our observations.⁹ The models confirm strong treatment effects in Stage I, even when accounting for end-game effects (column 2), project success (column 5), and opponents' previous choices (columns 3 and 6). Initiators are approximately 17 p.p. more likely to choose *Open* in Partial Reputation and 28 p.p. more likely in Full Reputation treatments compared to baseline. Partners are 25 p.p. more likely to choose *Keep* under Partial and 33 p.p. more likely under Full. The difference in the cooperation rates between the Full scheme and the Partial scheme is not statistically significant. These findings strongly support Hypothesis 1 regarding the positive effect of reputation schemes as monitoring devices. We summarize this result as follows:

Result 3. *Partial and Full reputation schemes are equally effective as monitoring devices. In Stage I, the likelihood of partners choosing the cooperative decision, Keep, is ordered as follows: Full = Partial > Baseline.*

In contrast to the strong treatment effects observed in Stage I, they essentially disappear in Stage II when reputation building ceases. Neither *Open* nor *Keep*

⁹The results remain robust with a Probit model.

	<i>Linear probability models</i>					
	Dependent: D_{Open}			Dependent: D_{Keep}		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.39** (0.05)	0.42** (0.05)	0.21** (0.03)	0.46** (0.07)	0.51** (0.09)	0.48** (0.09)
Partial	0.17* (0.08)	0.17* (0.08)	0.13* (0.06)	0.25** (0.09)	0.25** (0.09)	0.28* (0.11)
Full	0.28** (0.07)	0.28** (0.07)	0.18** (0.05)	0.32** (0.09)	0.33** (0.09)	0.38** (0.11)
Period 4&8		-0.12* (0.05)				
Success _t					-0.07 (0.07)	
Keep _{t-1}			0.54** (0.05)			
Open _{t-1}						-0.07 (0.08)
Stage II	-0.10 (0.06)	-0.08 (0.06)	0.01 (0.05)	0.03 (0.09)	0.06 (0.14)	-0.04 (0.10)
Partial × Stage II	-0.06 (0.09)	-0.06 (0.09)	-0.07 (0.07)	-0.28* (0.11)	-0.30** (0.11)	-0.36** (0.14)
Full × Stage II	-0.18 (0.10)	-0.18 (0.10)	-0.15 (0.08)	-0.41** (0.12)	-0.42** (0.12)	-0.50** (0.14)
Period 4&8 × Stage II		-0.08 (0.06)				
Success _t × Stage II					-0.03 (0.11)	
Keep _{t-1} × Stage II			-0.07 (0.08)			
Open _{t-1} × Stage II						0.22* (0.11)
Obs	762	760	664	340	338	279

Note: **p<0.05; ***p<0.01

Table 4: Linear probability regression results for the Initiator's *Open* decisions and the Partner's *Keep* decisions.

rates in the reputation treatments differ significantly from the baseline condition. This contradicts Hypothesis 2 on the positive value of reputation information. This finding highlights reputation schemes' greater effectiveness as a monitoring device over an informational one. The cross-stage comparison further reinforces this conclusion. Estimates in columns (1)-(2) and (4)-(5) reveal significant decreases in *Open* and *Keep* decisions from Stage I to II in both Partial and Full Reputation treatments. Specifically, initiators are approximately 14 p.p. less likely to choose Open

in the Partial and 26 p.p. less likely in the Full condition. Partners are about 24 p.p. less likely to choose Keep in the Partial and 34 p.p. less likely in the Full condition.¹⁰ Therefore, we establish the following result regarding the informational function of reputation schemes:

Result 4. *Partial and Full reputation schemes are not effective as informational devices. In Stage II, the likelihood of initiators choosing the cooperative decision, Open, is ordered as follows: Full = Partial = Baseline.*

Asymmetric Responses to Past Interactions. Our regression analysis also reveals a fundamental asymmetry in how initiators and partners respond to past interactions. This asymmetry becomes evident when we include historical interaction controls in models (3) and (6). Notably, the stage differences in cooperation that we observed earlier vanish for initiators when accounting for prior partners' choices (column 3) but strengthen for partners when controlling for prior initiators' behavior (column 6).¹¹ This pattern reveals distinct motivational mechanisms between initiators and partners. Initiators' cooperation decline from Stage I to Stage II is primarily explained by rational responses to partners' behavior rather than by the stage transition itself. In contrast, partners' cooperation drops in Stage II is not a response to how initiators treated them, but instead reflects pure strategic behavior. Quantitatively, initiators respond strongly to previous positive experiences. Those who experienced a cooperative partner (*Keep*) in the previous round are approximately 54 p.p. more likely to choose *Open* in the current round during Stage I and 47 p.p. more likely during Stage II averaged across treatments.¹² Both increases are significantly different from zero at the 5% level. In contrast, partners facing an initiator who chose *Open* in the previous round show no significant increase in *Keep* likelihood in either stage, confirming they are driven by forward-looking reputation concerns rather than backward-looking reciprocity.

To further quantify how initiators and partners respond differently to historical game outcomes across treatments, we extend the regression models in columns (3)

¹⁰These stage difference estimates are derived from regression models (2) and (5) and are all significantly different from zero at the 5% level. Models (1) and (4) yield similar estimates and significance.

¹¹The stage difference estimates are captured by the sum of the coefficients for the dummy for Stage II and its interaction with the Treatment dummy for Partial or Full. In column (3), the stage difference estimates are -0.06 for Partial and -0.14 for Full treatment. None of them is significantly different from zero at a level of 5%. In column (6), the stage difference estimates are -0.40 and -0.54. Both are significantly different from zero.

¹²Notice the benchmark group for the $Keep_{t-1}$ dummy includes cases where the partner chose *Leak* in $t - 1$ or the initiator chose *Close* in $t - 1$. Regressions separating these outcomes yield similar results to column (3).

and (6) by interacting previous-round decisions with treatment dummies and illustrate the average marginal effects in Figure 4. The Initiator-Partner asymmetry remains. Initiators consistently demonstrate strong responsiveness to prior interactions across all treatments; experiencing a cooperative partner in the previous round significantly increases the likelihood of choosing *Open* (indirect reciprocity) by 50-60 p.p. in most conditions. In contrast, partners exhibit minimal responsiveness to prior interactions. Experiencing a cooperative initiator in the previous round rarely impacts partners' inclination towards the cooperative action *Keep*.

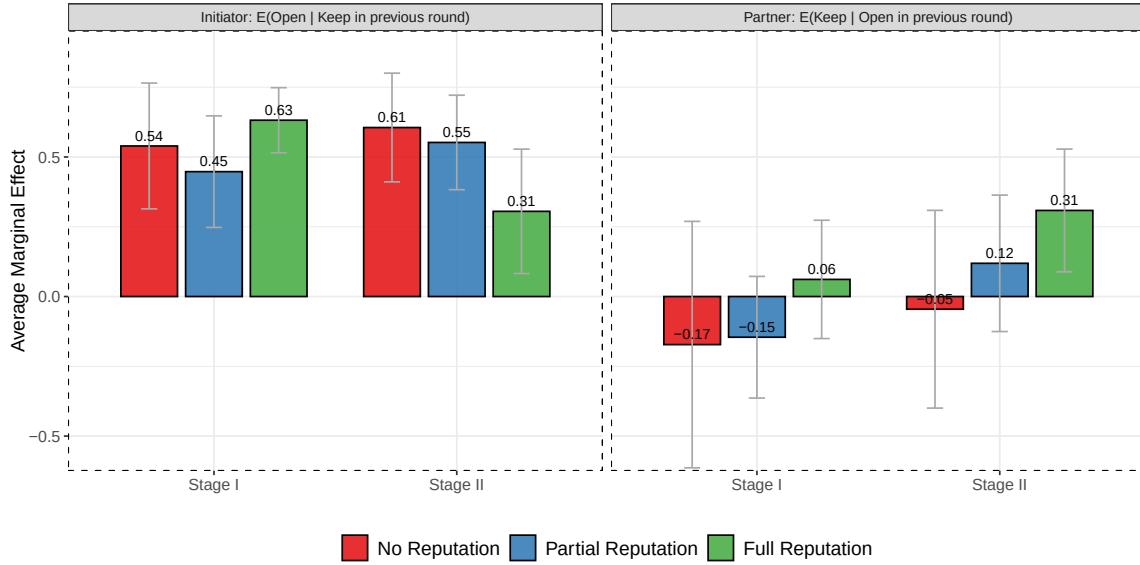


Figure 4: Average marginal effects of opponents' previous-round decisions on current cooperative choices by Initiators and Partners. Coefficients are obtained from linear probability models including Treatment, Stage, previous-round Decisions, and their interactions. Error bars represent the 95% confidence intervals. Compared to the No Reputation benchmark, only the Full Reputation treatment in Stage II shows significant differences for both regressions: Initiators ($p = 0.0472$) and Partners ($p = 0.09733$).

Result 5. *Initiators are significantly more likely to choose Open after experiencing cooperative partners in previous rounds across all treatments and stages. This responsiveness to partners' behavior explains the decline in Open rates between stages. In contrast, partners exhibit minimal responsiveness, with cooperative initiators having little impact on partners' Keep decisions.*

Interestingly, the Full Reputation treatment uniquely shows shifting patterns of responsiveness between stages. Initiators in this condition show a substantial decline in their positive response to cooperative partners, dropping from approximately 63 p.p. in Stage I to just 31 p.p. in Stage II, while partners emerge to become responsive, transitioning from unresponsiveness to a modest positive effect of 31 p.p. in Stage II. These divergent patterns reflect complementary mechanisms. For

initiators, the decline suggests a substitution effect where they rely less on their personal experience when comprehensive reputation information becomes available about their current partner. For partners, the emergence of responsiveness in Stage II likely stems from a crowding-out effect, whereby monitoring incentives in Stage I led them to prioritize strategic reputation building over genuine reciprocal responses, allowing authentic indirect reciprocity motives to surface only after monitoring ceased. These patterns highlight significant limitations in the reputation scheme's informational function that we explore further in subsequent analyses.

4.3 Quality of Reputation Information and Strategic Reputation Building

While reputation schemes theoretically provide valuable information to facilitate trust between initiators and partners, our findings reveal a critical limitation: partners strategically manipulate their reputation-building process, ultimately undermining the informational benefits that reputation schemes are designed to provide. This subsection examines how partners exploit the temporal structure of reputation recording and revealing, and why even high-quality reputation information fails to sustain cooperation.

The Paradox of High-Quality Reputation Information. We first examine the quality of the produced reputation information. Figure 5 illustrates the distribution of reputation information available to initiators across treatments, categorized by the *Keep/Open* ratio (times of *Keep* out of times of *Open*) that each partner achieved during Stage I.¹³ In both reputation treatments, more than 50% of initiators encounter partners with perfect reputations ($Keep/Open = 1$), indicating that these partners always chose *Keep* whenever given the opportunity in Stage I.¹⁴ This high concentration in the perfect reputation category further justifies our classification approach: partners with $Keep/Open = 1$ represent "Good Reputation" while those with $Keep/Open < 1$ represent "Bad Reputation."¹⁵

¹³The NoInfo cluster refers to situations where the Partner always encounters an Initiator who chooses Close, thus never having an opportunity to build a reputation.

¹⁴Notice that it holds for each round of Stage II, specifically, 17 out of 32 Initiator-Partner pairs in Partial Reputation and 20 out of 32 in Full Reputation fall into this "Good Reputation" category.

¹⁵We retain the NoInfo category and omit it when necessary due to the limited number of observations.

For comparison, we apply the same categorization standard and show initiators' experienced history information in Stage I in the bottom panel.¹⁶ Among the 32 Initiators in Stage I, only 8 encountered partners who always chose *Keep* during the 4 rounds in the Partial Treatment and 14 in the Full Treatment. The comparison reveals a striking pattern: the reputation information provided to initiators in Stage II is exceptionally good and significantly better than what they experienced through direct interactions in Stage I.¹⁷ This suggests that partners systematically built strong reputations during the reputation-recording phase, creating an apparent informational advantage that should theoretically facilitate trust in Stage II.

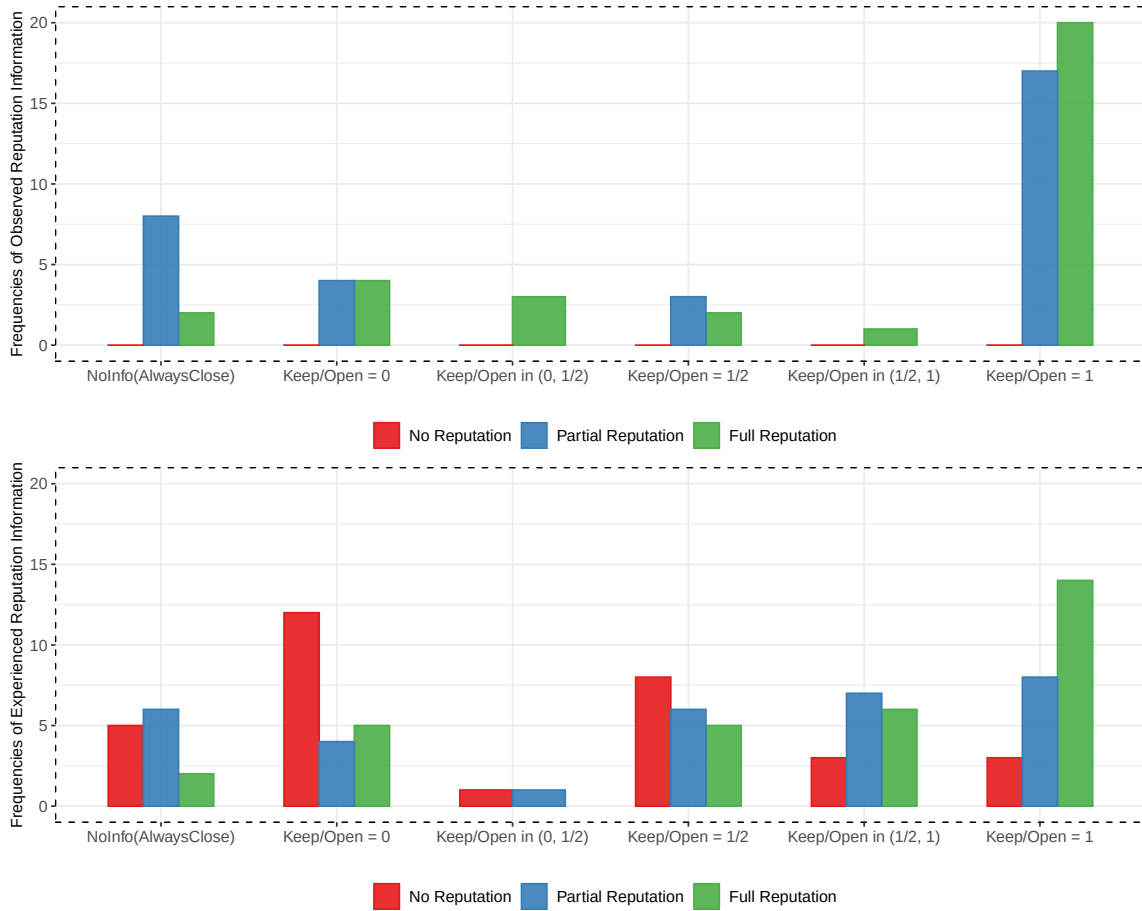


Figure 5: Distribution of Partner reputation profiles across treatments. The top panel shows reputation information available to Initiators in each round of Stage II, while the bottom panel displays the cumulative experienced history of Initiators with Partners after completing Stage I.

Result 6. *Partners build positive reputations: the reputation information available to initiators in Stage II is exceptionally good, with significantly more partners achieving perfect*

¹⁶This construction requires pooling observations across all four rounds in Stage I, thus representing the cumulative experience of Initiators after completing Stage I.

¹⁷The p values for the two-sample proportion test for the category *Keep/Open = 1* are 0.008 for Partial Treatment and 0.084 for Full Treatment.

reputations (Keep/Open = 1) than what initiators experienced through direct interactions in Stage I.

Despite the abundance of high-quality reputation information, Table 5 reveals a puzzling pattern that exposes the fundamental limitation of the information device. In Stage I, initiators encountered partners with good historical behavior (those who never met a partner chose Leak) maintained substantially higher Open rates of 75% in Partial and 82% in Full Reputation treatments. However, in Stage II, when initiators receive explicit reputation information—supposedly more relevant and informative than their own past experience—their Open rates toward partners with Good Reputation decline to just 56% in Partial and 46% in Full treatments. This unexpected drop suggests that good reputation information fails to effectively translate into sustained cooperative decisions by initiators, particularly under the Full scheme, where initiators appear more skeptical of reputation information.

	Treatment	Bad Reputation (Keep/Open < 1)	Good Reputation (Keep/Open = 1)
Stage II: Partial	Open%	25%	56%
	Keep%	0%	31%
Stage II: Full	Open%	20%	46%
	Keep%	2.5%	21%
Stage I: Partial	Open%	67%	75%
Stage I: Full	Open%	63%	82%

Table 5: Open and Keep rates by Good and Bad reputation categories. The last two rows provide benchmarks frequencies of *Open* in Stage I after the experienced good and bad reputation information.

More critically, the corresponding Keep rates among these supposedly trustworthy partners are remarkably low compared to their Open rates—just 31% in Partial and 21% in Full treatments. If partners maintained their established good reputations, we would expect Keep rates to exactly match Open rates. These significant discrepancies indicate that partners who cultivated perfect reputations in Stage I systematically exploit initiators’ trust in Stage II by frequently choosing *Leak* once monitoring ceases. This strategic manipulation of the reputation-building process, rather than the quality of reputation information itself, explains why the informational device fails to sustain cooperation between initiators and partners. We summarize the result as follows:

Result 7. *High-quality reputation information does not promote initiators' trust in Stage II: their Open rates towards partners with Good Reputation are lower compared to Stage I. Partners systematically exploit initiators' trust in Stage II: those with Good Reputation maintain Keep rates far below initiators' Open rates, revealing strategic reputation-building.*

Strategic Reputation Building and Exploitation of Good Reputation Figure 6 provides compelling evidence of strategic reputation building by partners and their continued exploitation. The top panel reveals a stark behavioral dichotomy. Partners with a good reputation maintain consistently high Keep rates of 60-80% throughout Stage I, actively building valuable reputations while being monitored. However, upon entering Stage II when monitoring ceases, these partners dramatically shift their behavior. Keep rates decline substantially from approximately 70% to below 30% in both treatments, though more immediately and sharply in the Full treatment. This precipitous decline directly contrasts with the stable behavior of partners with a bad reputation, who maintain consistently low Keep rates across both stages. By the final period, good reputation partners' keep rates converge with those of bad reputation partners. This temporal pattern demonstrates that partners who built good reputations in Stage I were not intrinsically more cooperative, but rather strategically patient—investing in reputation in Stage I to maximize exploitation opportunities in Stage II.

The bottom panel of Figure 6 illustrates initiators' responses to partners' good and bad reputation information. The horizontal lines in Stage I represent counterfactual baseline Open rates—how the same cohorts of initiators behaved before receiving reputation information. Initiators who would later encounter partners with good and bad reputations exhibit similar Open rates, around 50% in Partial and 65% in Full treatments during Stage I. Upon entering Stage II, initiators in Partial exhibit an initial trust in good reputation partners, reflected by an increase in their Open rates. However, initiators in the Full treatment partially anticipate strategic manipulation and respond by immediately reducing their Open rates. As both groups experience systematic exploitation by supposedly trustworthy partners, their Open rates steadily decline throughout Stage II. Conversely, initiators facing partners with a bad reputation show consistently lower Open rates throughout. This pattern of initial trust followed by learned skepticism reveals that while reputation information may initially enhance initiators' trust decisions (showing a limited informational role), strategic reputation building by partners ultimately undermines its effectiveness.

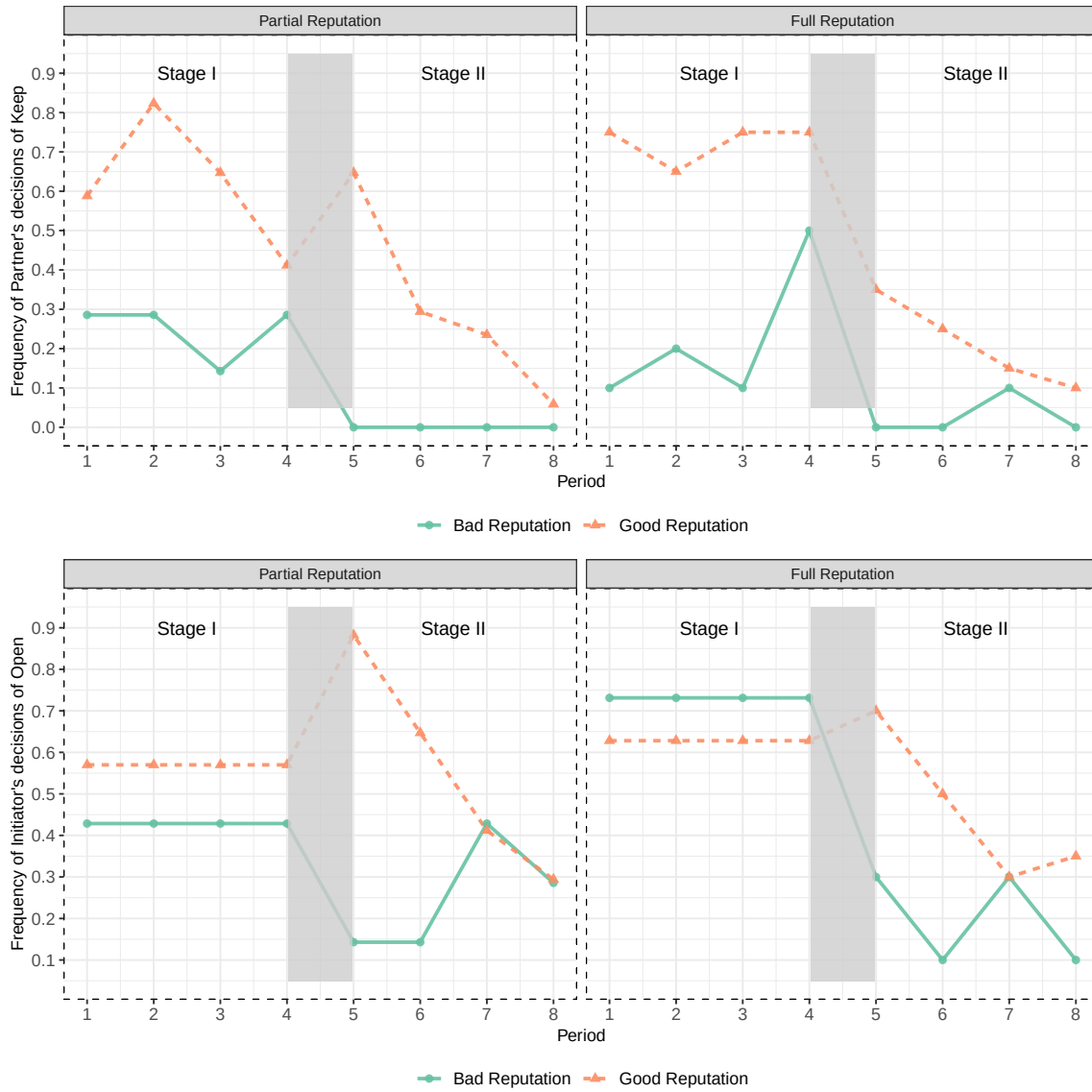


Figure 6: Keep rates and Open rates over time by Good and Bad Reputation categories. The top panel displays the frequencies of *Keep* decisions by Partners over time, categorized by their Good and Bad reputation status established during Stage I. The bottom panel displays the *Open* decisions in response to Partner reputation. Horizontal lines in Stage I represent counterfactual baseline Open rates for the same cohorts of Initiators before receiving reputation information.

Result 8. *The informational function of reputation schemes is limited in promoting cooperation due to partners' strategic exploitation: partners deliberately build a Good Reputation during Stage I while being monitored, but immediately exploit initiators by reducing cooperation substantially in Stage II (especially under the Full scheme). While initiators overall respond positively to Good Reputation information (more pronounced in Partial treatment), they gradually reduce trust as exploitation occurs.*

These temporal patterns are further confirmed through regression analysis.

Figure 7 presents the average marginal effects of Good Reputation on cooperative decisions in Stage II, derived from linear probability models detailed in Appendix B.2, Table 8. The left panel quantifies what Figure 6 illustrates about initiators: they consistently respond positively to Good Reputation information, with Good Reputation increasing Open likelihood by approximately 30 p.p. in Partial and 20 p.p. in Full treatment.¹⁸ The more substantial effect in Partial than Full treatment confirms that reputation schemes retain informational value, though this value diminishes when initiators can better anticipate strategic manipulation in the Full Reputation treatment. The right panel statistically confirms the exploitation pattern observed in the top panel of Figure 6: while good reputation partners show marginal residual cooperation in Partial treatment, those in Full treatment exhibit Keep rates statistically indistinguishable from partners with unfavorable reputation information, confirming strategic reputation exploitation in the Full Reputation treatment.

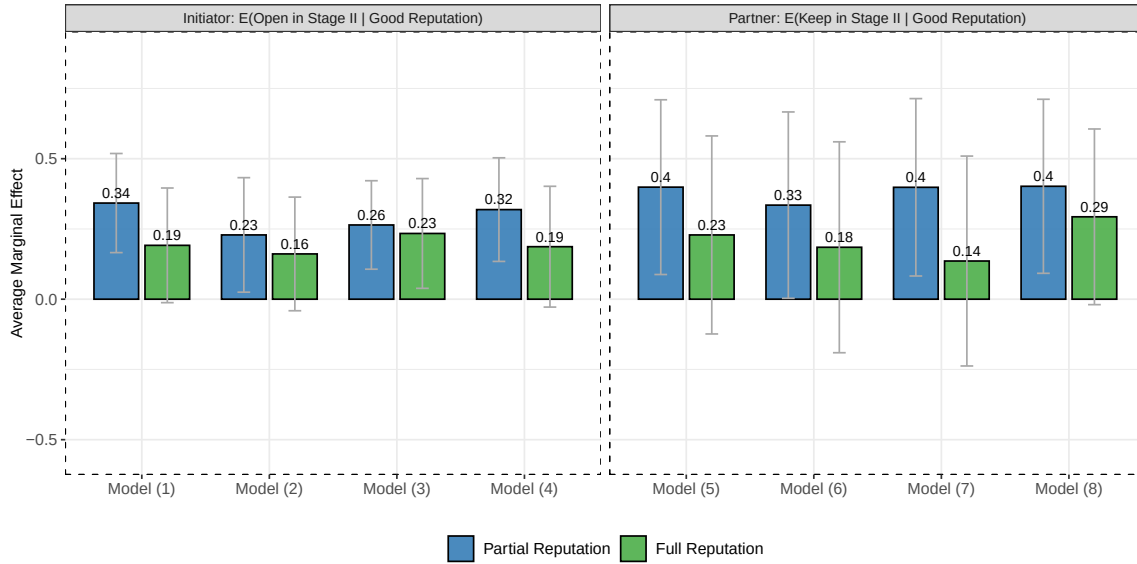


Figure 7: Histogram of the average marginal effects of Good Reputation on Initiators' and Partners' cooperative decisions in Stage II. The coefficients for Partial and Full are obtained from their interactions with the Good Reputation dummy and control for Period 5 (Models 2 and 6), previous round decisions (Models 3 and 7), and cumulative cooperation counts in Stage I (Models 4 and 8). The error bars represent the 95% confidence intervals.

Our regression analysis also confirms the previous finding (Result 5) on asymmetric responses to past interactions between initiators and partners. Initiators become approximately 50% more likely to choose *Open* in Partial and 34% more

¹⁸The effect sizes remain robust across specifications controlling for first-period exposure (Model 2), previous-round Partner action (Model 3), and partners' cumulative cooperative interactions measured by total number of *Keep* in Stage I (Model 4).

likely in Full after experiencing cooperative partners.¹⁹ The marginal effect of the number of *Keep* experienced by Initiators in Stage I (model (4)) is significant in Partial but not Full treatment. This indicates that providing reputation information complements, rather than crowds out, indirect reciprocity induced by experienced interaction. In contrast, estimates from models (6) and (7) show that Partners remain unresponsive to prior Initiator actions or accumulated cooperative experiences from Stage I²⁰, confirming their strategic orientation toward exploitation rather than genuine cooperation. While both partial and full reputation schemes generate Good Reputation information and thus encourage cooperation from initiators, they fail to induce lasting cooperation from Partners.

In summary, we find that reputation schemes function effectively as monitoring devices but create exploitable vulnerabilities once monitoring ceases. Sophisticated partners systematically exploit this temporal structure, building favorable reputations during monitored phases only to abandon cooperation afterward, particularly in the Full Reputation treatment. This strategic behavior limits the informational function of reputation schemes and undermines their long-term effectiveness in promoting sustained cooperation.

5 Conclusion

This paper deepens our understanding of the nuanced mechanics and operation of reputation schemes in fostering collaboration amid the trust dilemmas posed by one-sided incentive problems. While reputation schemes are widely believed to foster cooperation through both monitoring and information provision, their relative contributions to this process remain unclear. We experimentally disentangle these dual roles through a novel two-stage design that separates the reputation-recording phase (Stage I) from the reputation-revealing phase (Stage II) in a repeatedly played variation of the trust game between initiators and partners. By comparing three treatment conditions (No Reputation, Partial Reputation, and Full Reputation), we demonstrate that reputation schemes promote cooperation primarily as monitoring devices rather than informational ones—cooperative behavior emerges strongly during the reputation-building phase but collapses when monitoring ceases, despite the availability of high-quality reputation information

¹⁹The two numbers are statistically significantly different from zero at a 5% level based on estimates from model (3) in Table 8.

²⁰Hypothesis test for estimates in models (6) and (7) show that none of the marginal effects is significantly different from zero.

for initiators.

We find that both Full and Partial Reputation schemes significantly promote cooperative behavior, particularly when the monitoring function is present. In Stage I, more than half of the partners achieve perfect cooperation records. However, in Stage II where the monitoring function is absent, partners increasingly defect, and cooperation levels converge to those observed in the No Reputation treatment. In both partial and full treatments, a significant number of partners exploit initiators' trust by building perfect cooperation records in Stage I and then defecting in Stage II. This exploitation is most pronounced under the Full Reputation scheme, suggesting that partners strategically anticipate stronger informational effects when more reputation information is available.

Our findings have important theoretical implications. Traditional reputation models often treat monitoring and information provision as complementary aspects of a unified mechanism. By isolating these functions, our work reveals that agents respond differently to each component. The monitoring role aligns with the forward-looking reasoning of reputation, where agents strategically behave cooperatively to maintain future opportunities. However, the limited effectiveness of reputation's informational role suggests that purely backward-looking reasoning overestimates the value of reputation information when agents can anticipate changes in the intensity of monitoring.

Our temporal distinction also advances the literature beyond the traditional focus on reputation's effectiveness in fixed environments to consider how agents strategically adapt their behavior to changes in reputation system structure. While previous research has examined reputation building in continuous monitoring environments or studied the value of reputation information in isolation, our work bridges these streams by demonstrating how these components interact across time. The strategic exploitation of reputation systems we observe highlights the importance of considering dynamic behavioral responses when designing reputation schemes.

Our findings have practical implications for efficient organizational design and effective reputation scheme implementation, particularly in contexts with asymmetric information and incentives (Chen, Cramton, List, & Ockenfels, 2021; Delarocas, 2010). First, effective reputation schemes should maintain continuous monitoring rather than relying on accumulated historical information. The dramatic decline in cooperation when monitoring ceases suggests that maintaining

active recording of current behaviors is critical for sustaining cooperative outcomes. Second, partial reputation schemes may offer nearly equivalent benefits to comprehensive monitoring while requiring fewer resources, as the mere threat of potential monitoring appears sufficient to induce cooperative behavior. Third, organizations should anticipate strategic exploitation by implementing reputation decay mechanisms that give recent behaviors greater weight than past actions.

Several limitations of our study suggest directions for future research. Our laboratory experiment uses discrete stages with fixed reputation schemes. Future work could examine more dynamic reputation systems with gradual transitions between monitoring and informational intensities, and explore how agents respond to more complex reputation metrics. Field studies in actual open innovation contexts could validate our findings and identify additional factors that might enhance the effectiveness of reputation's informational role in practice. Additionally, research on mechanisms to counter strategic reputation building and exploitation would complement our findings on this vulnerability.

In conclusion, reputation mechanisms function primarily as monitoring devices that strategically transform opportunistic agents into cooperative ones while monitoring persists. When designing reputation systems, organizations should focus on maintaining forward-looking incentives throughout interactions rather than assuming that reputation information alone will sustain cooperation. By understanding the dual roles of reputation and their asymmetric effects, organizations can better leverage these mechanisms to facilitate productive collaboration while mitigating opportunistic exploitation in open innovation and other trust dilemma contexts.

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A Experiment instructions for Partial Reputation treatment)

Introduction

This experiment is about innovation decisions. The experiment is not complicated. Please read the instructions carefully and make your decisions with a good understanding of this instruction. When the experiment ends, we will privately pay you your earning from the experiment in cash.

Reading this instruction is very important to helps you understanding this experiment. We will also ask you to answer a few simple questions after reading instructions. If you don't understand the whole content of the experiment correctly, you may lose the opportunity to get paid in this experiment.

During the entire experiment, please do not talk to the other participants. Please turn off your phone and only use your computer to proceed your decision. If you have any questions, please raise your hand, and our experimenter will answer your questions.

You will participant in the decisions about innovation. There are two roles in the experiment: Company and External Developer. You will be randomly assigned to play one of the roles, and your role will be fixed throughout the experiment. The experiment consists of two stages. The first stage is from Round 1 to 4, where you will pair with four participants who play another role, respectively. The second stage is from Round 5 to 8, where you will pair with another four participants, respectively. In each round, you will make the same Decisions about technology innovation. The decisions do not differ between two stages.

Your profit for each round will be accumulated and paid to you in cash after the experiment is over. The currency unit used in the experiment is *RMB* yuan

How to make decisions

At the beginning of each round, the company role player first chooses the innovation method: 1. Independent innovation; or 2. open innovation.

1. Independent innovation

In this case, the external developer will not participate in this innovation process. The innovation project will have a success with a probability of 30%

and a failure with a probability of 70%. If the project successes, the company obtains 10 yuan; Otherwise, the company obtains 4 yuan. The external developer obtains 4 yuan regardless success or failure.

2. Open innovation In this case, the external developer will take part in the innovation process, where the company shares the external developer production information. In this process, the project will success with a probability of 70% probability and will fail with a probability of 30%. After the innovation outcome realized, the external developer chooses whether to protect the company's production information.
 - a. If the external developer chooses to "keep" production information, both parties will get 10 yuan in case of success, and 4 yuan in case of failure.
 - b. If the external developer chooses to "leak", the company will get 2 yuan and the external developer will get 14 yuan in case of success; the company will get 0 yuan and the external developer will get 6 yuan in case of failure.

How the innovation result is determined

In each round, after the company chooses the innovation method, the company player draws a ball from a black bag with a number of red and blue balls. Draw out the red ball to indicate successful innovation, and draw out the blue ball to indicate innovation failure. There are 3 red balls and 7 blue balls in the black bag if the company chooses independent innovation; and 7 red and 3 blue balls in the bag if the company chooses open innovation.

Reputation system

In each round, the experiment has two stages as described above.

In the first stage, there are four rounds, and the decision of the external developer will be recorded. The information format is (innovation method, innovation result, decision of the external developer). If the innovation method is independent, the external developer no needs to make decision and it is recorded as (independent, -, -). At this stage, the company cannot see any historical information about the counterpart external developer.

The second stage still has four rounds. The company will be able to see the two rounds' previous decision results of the counterpart external developer (which two

rounds are randomly determined). The decision of the external developer will not be recorded in this stage.

It should be noted that in the eight rounds of the experiment, you were matched with eight different participants.

Throughout the experiment, you can review your decision and payoff results in all previous rounds.

B Additional analyses

B.1 Treatment effects after project success and failure

Game Outcome	No Reputation	Partial Reputation	Full Reputation
(Close, —, —)	66%	52%**	48%**
(Open, Success, Keep)	12%	18%	24%**
(Open, Fail, Keep)	4%	11%**	10%*
(Open, Success, Leak)	12%	15%	16%
(Open, Fail, Leak)	6%	4%	4%

Table 6: Frequencies of five game outcomes across three reputation schemes. Statistical significance is evaluated relative to the No Reputation condition. Comparisons between Partial Reputation and Full Reputation reveal no significant differences in any of the outcomes. Statistical tests employed are two-sample proportion test with * $p < 0.05$ and ** $p < 0.01$.

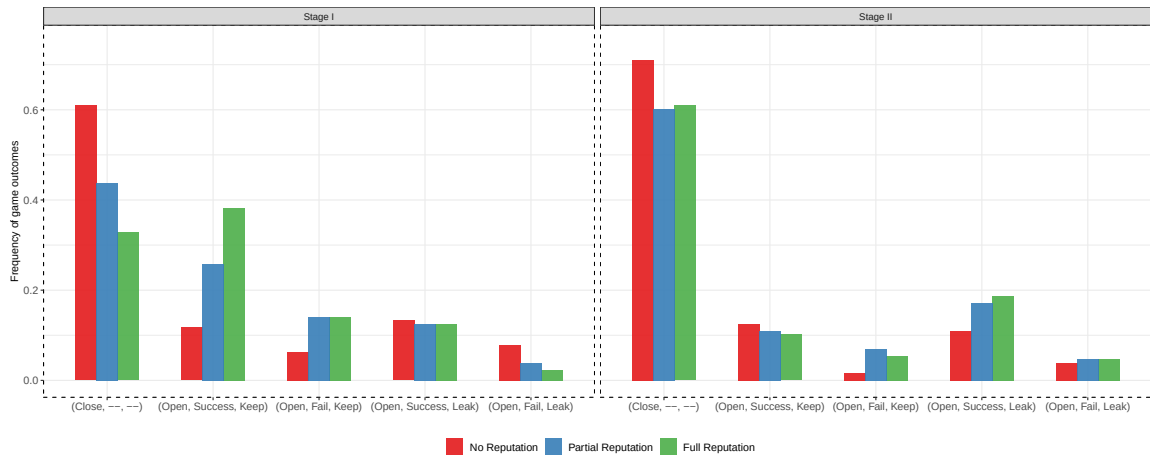


Figure 8: Histogram of Initiator-Partner game outcomes, including an additional indicator for project success or failure. It is important to note that the Partner's acts after learning whether the project succeeded or failed.

Decision: <i>Keep</i>		No Reputation	Partial Reputation	Full Reputation
Success	Stage I	15 (47%)	33 (67%)	49 (75%)**
	Stage II	16 (53%)	14 (39%)	13 (35%)
	Stage I vs. II		**	**
Failure	Stage I	8 (44%)	18 (78%)*	18 (86%)**
	Stage II	2 (29%)	9 (60%)	7 (54%)
	Stage I vs. II			*

Table 7: Frequencies of the Partner's *Keep* decision after project success and failure. Statistical significance is assessed relative to the No Reputation condition. There is no significant differences in any comparisons between Partial and Full Reputation conditions. Both the two-sample *t*-test and the Wilcoxon rank sum test produce consistent significance results, with ** $p < 0.05$ and *** $p < 0.01$.

B.2 Additional regressions for the effects of reputation information on Open and Keep decisions

	<i>Linear probability models</i>							
	Dependent: D_{Open}				Dependent: D_{Keep}			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Constant	0.22** (0.06)	0.22** (0.07)	0.13** (0.05)	0.07 (0.06)	0.15 (0.11)	0.10 (0.10)	0.15 (0.13)	0.18 (0.21)
GoodRepute	0.34** (0.09)	0.23* (0.10)	0.26** (0.08)	0.32** (0.09)	0.40* (0.16)	0.33 (0.17)	0.40* (0.16)	0.40* (0.16)
Full	0.05 (0.09)	0.00 (0.09)	0.01 (0.09)	0.22 (0.12)	0.08 (0.18)	0.15 (0.17)	-0.01 (0.20)	0.17 (0.29)
GoodRepute $\times$ Full	-0.15 (0.14)	-0.07 (0.15)	-0.03 (0.13)	-0.13 (0.14)	-0.17 (0.24)	-0.15 (0.26)	-0.26 (0.25)	-0.11 (0.22)
Period5		-0.02 (0.11)				0.23 (0.29)		
GoodRepute $\times$ Period5		0.45** (0.16)				0.07 (0.32)		
Period5 $\times$ Full		0.22 (0.18)				-0.28 (0.32)		
GoodRepute $\times$ Period5 $\times$ Full		-0.33 (0.24)				0.05 (0.38)		
Keep $_{t-1}$			0.50** (0.08)					
Keep $_{t-1}$ $\times$ Full			-0.16 (0.13)					
#Keep(StageI)				0.10** (0.03)				
#Keep(StageI) $\times$ Full				-0.11* (0.05)				
Open $_{t-1}$							0.001 (0.14)	
Open $_{t-1}$ $\times$ Full							0.27 (0.21)	
#Open(StageI)								-0.01 (0.06)
#Open(StageI) $\times$ Full								-0.05 (0.09)
Obs	252	246	250	250	97	93	95	93
Note:						*p<0.05; **p<0.01		

Table 8: The effect of reputation information (categorization) on Initiator's *Open* decisions and the Partner's *Keep* decisions in Stage II through linear probability regression models. The benchmark group contains both the *BadReputation* and *NoInfo(AlwaysClose)* categories due to the limitations on observations. There are only two Partners fall into the category of *NoInfo(AlwaysClose)* in the Full Reputation treatment.